

Rajasthan Board Class 12**Subject – Physics****Time:** $3\frac{1}{4}$ Hours**M.M:** 56**Question 1:****Define 'drift velocity' of electrons in a conductor****Sol:**

The velocity with which the free electrons get drifted towards the +ve terminal under the influence of an external applied field is known as drift velocity.

Question 2:**Write the name of fields produced by a moving charged particle.****Sol:**

Electric and magnetic fields are produced by a moving of charged particle.

Question 3:**Define Curie temperature in magnetism.****Sol:**

The transition temperature above which ferromagnetic substances start behaving like paramagnetic substances is known as Curie temperature T_c .

Question 4:**Why self-induction is called inertia of electricity?****Sol:**

Self-induction is the property which maintains the magnetic flux linked with it and opposes any changes in the flux inducing current in it.

Question 5:

A charged particle oscillates about its equilibrium position with a frequency of 100MHz . What is the frequency of electromagnetic waves produced by the

oscillator?

Sol:

The frequency of electromagnetic waves will be 100MHz . Because the frequency of electromagnetic waves produced by the oscillator is always same as the frequency with which a charged particle oscillating about its mean position

Question 6:

"Light added to light can produce darkness". Write the name of this phenomenon.

Sol:

In destructive interference or destructive superposition lights are superimposed in opposite phase then darkness is produced.

Question 7:

Write the definition of refractive index of a medium.

Sol:

Refractive index can be defined as the ratio of the speed of light in one medium to that in another medium.

Question 8:

Write Heisenberg's uncertainty principle

Sol:

Heisenberg's uncertainty principle, states that it is not possible to measure both the position and momentum of an electron at the same time exactly. There is always some uncertainty in the specification of position and momentum.

Question 9:

An electron, an alpha (α) particle and a proton have same kinetic energies. Which one of these particles has largest de Broglie wavelength?

Sol:

We have the de Broglie wavelength is given as $\lambda = \frac{h}{\sqrt{mE}}$

Where, h is the Planck's constant, m is mass of particle and E is the kinetic energy of the particle.

Therefore,

$$\lambda \propto \frac{h}{\sqrt{m}}$$

Because, de Broglie wavelength is inversely proportional to the mass of the particle.

Question 10:

Write the Rydberg formula for the spectrum of the hydrogen atom.

Sol:

The Rydberg formula for the spectrum of the hydrogen atom is given below:

$$\frac{1}{\lambda} = R \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

Where, R is the Rydberg's constant

n_1 and n_2 are integers $n_1 < n_2$.

Question 12:

What is broadcast communication?

Sol:

A large number of receivers corresponding to a single transmitter is known as broadcast communication. Like radio and television.

Question 13:

Which waves normally broadcast the frequencies in ultra high frequency (UHF) range?

Sol:

Space waves.

Question 14:

A parallel plate capacitor with air has a capacity of 8 PF. Calculate the capacity of the capacitor if the distance between the plates is halved and the space between them is fully filled with a substance of dielectric constant 5.

Sol:

We have,

The capacity of a parallel plate capacitor is with air is $\frac{A\epsilon_0}{d}$ equal to 8PF.

Now, from the question we get that distance between the plates is halved and the space between

them is fully filled with a substance of dielectric constant 5.

Therefore,

The capacity of the capacitor is

$$\begin{aligned} & \frac{kA\epsilon_0}{\frac{d}{2}} \\ &= 10 \left(\frac{A\epsilon_0}{d} \right) \\ &= 10 \times 8 \\ &= 80PF \end{aligned}$$

Question 15:

What is Ohm's law? Write any two limitations of it.

Sol:

The electric current flowing through a conductor is directly proportional to the potential difference across the two ends of the conductor provided physical conditions temperature is known as Ohm's law.

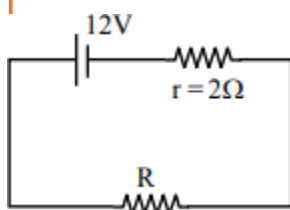
Limitations of Ohm's are as follows.

- (i) Temperature should remain the same.
- (ii) Law is valid only for conductors, not for semiconductor electrolytes etc.

Question 16:

A battery of electromotive force 12V and internal resistance 2 ohm is connected to a resistor. If the current of 0.5A is flowing through the circuit then calculate the resistance of resistor. If the circuit is closed then what will be terminal voltage of the cell?

Sol:

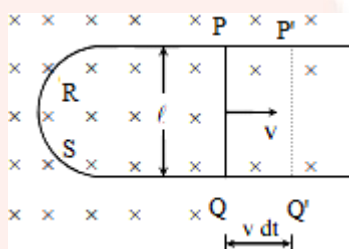


$$\begin{aligned}\therefore E &= I(r + R) \\ \Rightarrow 12 &= 0.5(2 + R) \\ \Rightarrow 24 &= 2 + R \\ \Rightarrow R &= 22\Omega\end{aligned}$$

And the terminal voltage, V is given by

$$\begin{aligned}E - Ir \\ = 12 - (0.5)(2) \\ = 11\text{volt}\end{aligned}$$

Question 17:



A conducting rod of length ' l ' is moving with constant linear speed ' v ' in a uniform magnetic field ' B '. This arrangement is mutually perpendicular. Obtain the expression of motional electromotive force.

Sol:

Let us consider a straight conductor PQ moving with velocity v towards right in a U-shaped conductor placed in a uniform magnetic field perpendicular to the plane of paper. When the conductor slides, the area of the circuit changes from $PQRS$ to $P'Q'RS$ in time dt .

The change in magnetic flux $d\phi_B$ is equal to the product of magnetic field and change in area.

$$\begin{aligned}\therefore d\phi_B &= B \times \text{Area}(PQ'P') \\ &= Blvdt\end{aligned}$$

The electromotive force is the rate of change in magnetic flux.

$$\begin{aligned}\therefore E &= \frac{d\phi_B}{dt} \\ &= \frac{Blvdt}{dt} \\ &= Blv\end{aligned}$$

Question 18:

Write the name of any four waves (radiations) produced in electromagnetic spectrum.

Sol:

The four waves produced in electromagnetic spectrum are Gamma rays, X-rays, Ultraviolet rays, Infrared rays and microwaves.

Question 19:

The focal lengths of an objective lens and eyepiece are 192 cm and 8 cm respectively in a small telescope. Calculate its magnifying power and the separation between the two lenses.

Sol:

The focal length of an object f_o and eyepiece f_e are 192 cm and 8 cm respectively.

And the magnifying power m_p of the telescope is equal to the ratio of focal length of eyepiece and objective.

$$\begin{aligned}\therefore m_p &= \frac{f_o}{f_e} \\ &= \frac{192}{8} \\ &= 24\end{aligned}$$

Now, the separation between the two lenses is equal to the sum of the focal lengths

$$\begin{aligned}\therefore L &= f_o + f_e \\ &= 192 + 8 \\ &= 200\text{cm}\end{aligned}$$

Question 20:

What is photo electric effect? On which two factors the photo electric current depends.

Sol:

The process of emission of electron by a certain substance, when it is exposed to radiation of suitable frequencies is known as photo electric effect. And the emitted electrons are known as photoelectrons.

It depends on,

- (i) Intensity of incident light
- (ii) Potential of collector with respect to emitter

Question 21:

What are optoelectronic junction devices? Write any two names of optoelectronic junction devices.

Sol:

Electronic devices and systems which detect control and produce light is known as optoelectronic junction devices

Two names of optoelectronic junction devices are as follows:

- (i) Light emitting diodes (LED)
- (ii) Photovoltaic devices

OR

What are logic gates? Write two names of universal logic gates.

Sol:

The fundamental building blocks of digital integrated circuits which perform basic logical functions are known as logic gates.

A universal gate is a gate which can implement any Boolean function without need to use any other gate type. NAND and NOR gates.

Question 22:

Write two limitations of Bohr's model.

Sol:

The two limitations of Bohr's model are following

- (i) The spectrum of only simple atoms like 'H' is explained in Bohr's theory. For atoms with more than one electron, this theory is inadequate.
- (ii) In this theory, the wave nature of electrons is not explained

Question 23:

a) Define the activity of radioactive substances. Write its S.I. unit.

Sol:

The rate of decay of radioactive substance is known as the rate of decay of radioactive substance. Its SI unit is Bacquerel(dps)

b) "The half-life of $^{14}\text{C}_6$ is 5700 years." What does it mean?

Sol:

The half-life of $^{14}\text{C}_6$ is 5700 years. This shows that the C^{14} nuclei of any sample will decay during 5700 years.

Question 24:

Distinguish between modulation and demodulation

Sol:

The difference between modulation and demodulation are following:

MODULATION	DEMODULATION
Modulation is the process of varying the Parameter of the carrier signal according to message bearing signal.	Demodulation is the process by which message signal is extracted from the modulated wave.
The circuit needed for modulation is Modulator	The circuit needed for demodulation is Demodulator/Detector

Question 25:

Describe any two energy losses occurring in transformers. How these can be minimized? Why electric power is transmitted at high voltage up to large distances?

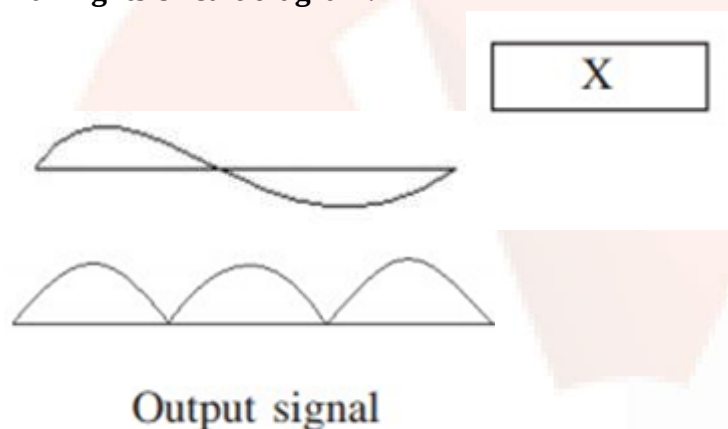
Sol:

Two energy loosing occurring in transformers are as follows:

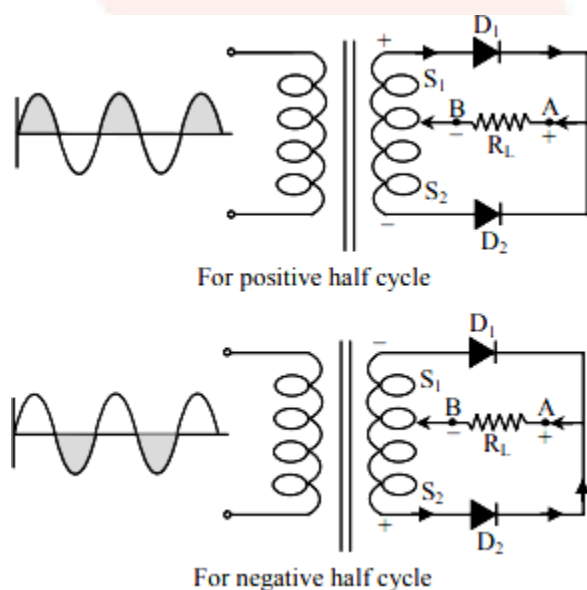
- (i) **Flux Leakage:** All the flux due to primary does not pass through the secondary due to poor design of the core or the air gaps in the core. It can be reduced by winding the primary and secondary coils one over the other.
- (ii) **Eddy current:** Alternating magnetic flux induces eddy currents in the iron core and causes heating. It can be reduced by using a laminated core. Current is minimized which decreases power loss in long distance transmission at high voltage. It is caused due to the electric power is transmitted at high voltage up to large distances.

Question 26:

Write the name of device 'X' in the following given diagram. Explain its working making its circuit diagram.



Sol:

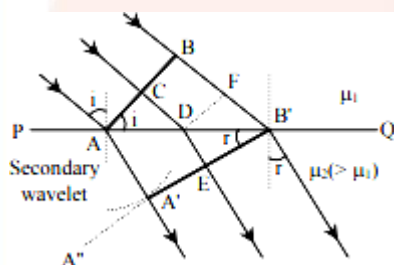


The alternating input signal is fed to the primary P_1P_2 of a transformer. And the output signal appears across the load resistances R_L . During the positive half of the input signal, suppose P_1 and P_2 are negative and positive respectively. It means that S_1 and S_2 are positive and negative respectively. So, the diode D_1 is forward biased and the diode D_2 is reverse-biased. And the flow of current in the load resistance R_L is from A to B. During negative half of the input signal, S_1 and S_2 will be negative and positive respectively. Such that D_1 will be reverse-biased and D_2 will be forward biased. So, it is clear that the flow of current is independent from the input signals.

Question 27:

What is wave front? Explain laws of refraction of light on the bases of Huygens wave theory.

Sol:



The locus of points having the same phase of oscillation is known as wave front.

Formation of refracted wave front:

Let us suppose a plane wave front AB be incident on a refracting plane surface PQ separating a rarer medium of refractive index μ_1 from a denser medium of refractive index $\mu_2 (> \mu_1)$. During this time the disturbance from B reaches B' , the disturbance from A must have traveled a distance $AA' = c_2 t$, where c_2 is the velocity of light in denser medium and t is the time taken by the disturbance to travel from B to B' .

$$\sin i = \frac{BB'}{AB'} \text{ and } \sin r = \frac{AA'}{AB'}$$

$$\therefore \frac{\sin i}{\sin r} = \frac{BB'}{AA'}$$

$$= \frac{c_1 t}{c_2 t}$$

$$= \frac{c_1}{c_2}$$

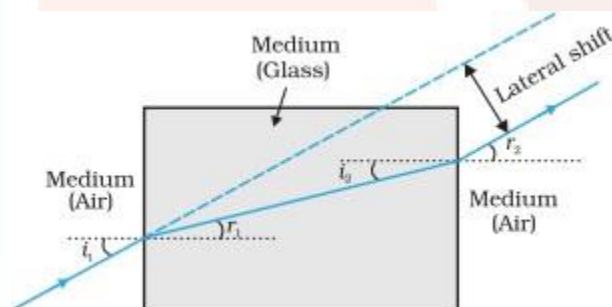
$$= \text{const } \tan t.$$

OR

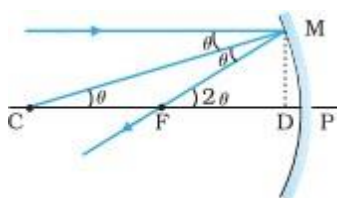
What is lateral shift? Establish relation between focal length and radius of curvature of a mirror.

Sol:

The perpendicular distance between the incident ray and the emergent ray is known as lateral shift.



Let us consider a ray parallel to the principal axis striking the mirror at M. Then CM will be perpendicular to the mirror at M. Let θ be the angle of incidence, and MD be the perpendicular from M on the principal axis.



$$\tan \theta = \frac{MD}{CD} E \text{ and } \tan 2\theta = \frac{MD}{FD} \dots (1)$$

For small θ , $\tan \theta = \tan 2\theta = \theta$

$$\therefore \frac{MD}{CD} = \theta, 2\theta = \frac{MD}{FD}$$

$$\Rightarrow FD = \frac{CD}{2} \dots (2)$$

Now, for the small θ point D is very close to P

So,

$$FD = f \text{ and } CD = R$$

From eq(2) we get,

$$f = \frac{R}{2}$$

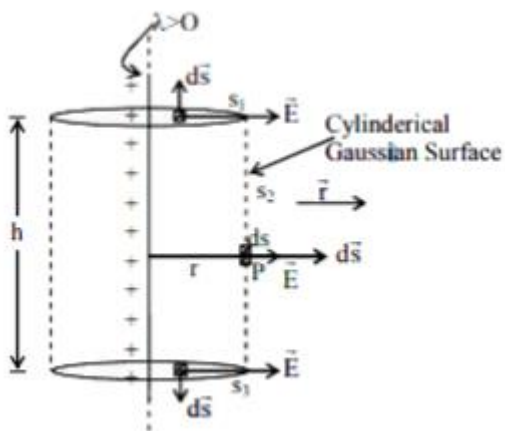
Question 28:

State Gauss law in electrostatics. Derive an expression of electric field due to an infinitely long straight uniformly charged wire. Draw necessary diagram.

Sol:

The surface integral of the electric field intensity over any closed hypothetical surface in free space is equal to $\frac{1}{\epsilon_0}$ times the net charge enclosed within the surface. ϵ_0 is the absolute permittivity.

$$\begin{aligned} \phi_E &= \oint \vec{E} \cdot d\vec{s} \\ &= \frac{1}{\epsilon_0} \sum_{i=1}^n q_i \end{aligned}$$



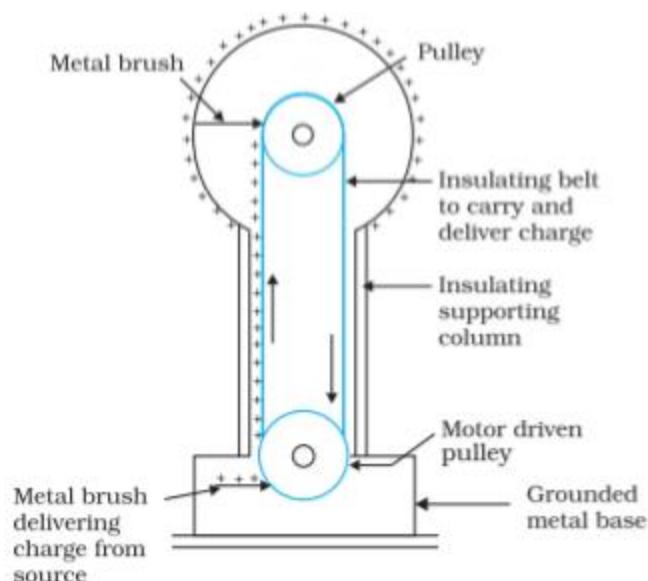
And the cylindrical Gaussian surface can be divided into three parts I II III

$$\begin{aligned}
 \oint \vec{E} \cdot d\vec{S} &= \frac{1}{\epsilon_0} q \\
 \Rightarrow \int_I \vec{E} \cdot d\vec{S} + \int_{II} \vec{E} \cdot d\vec{S} + \int_{III} \vec{E} \cdot d\vec{S} &= \frac{h\lambda}{\epsilon_0} \\
 \Rightarrow \int E \cdot ds \cos 0^\circ + \int E \cdot ds \cos 90^\circ + \int E \cdot ds \cos 90^\circ &= \frac{h\lambda}{\epsilon_0} \\
 \Rightarrow E \int dS &= \frac{h\lambda}{\epsilon_0} \\
 \Rightarrow E(2\pi rh) &= \frac{h\lambda}{\epsilon_0} \\
 \Rightarrow E &= \frac{\lambda}{2\pi r \epsilon_0}
 \end{aligned}$$

OR

Describe the construction and working of Van De Graff generator making the labeled diagram.

Sol:



It is a machine which is capable of building up potential difference of few million volts and field close to breakdown field of air which is about $3 \times 10^6 \text{ V/m}$. It consists of a large spherical conducting shell (of few meters radius) is supported at a height several meters above the ground on an insulating column. A long narrow endless belt insulating material, like rubber or silk, is wound around two pulleys—one at ground level, one at the center of the shell. This belt is kept continuously moving by a motor driving the lower pulley. It continuously carries positive charge, sprayed on to it by a brush at ground level, to the top then it transfers its positive charge to another conducting brush connected to the large shell.

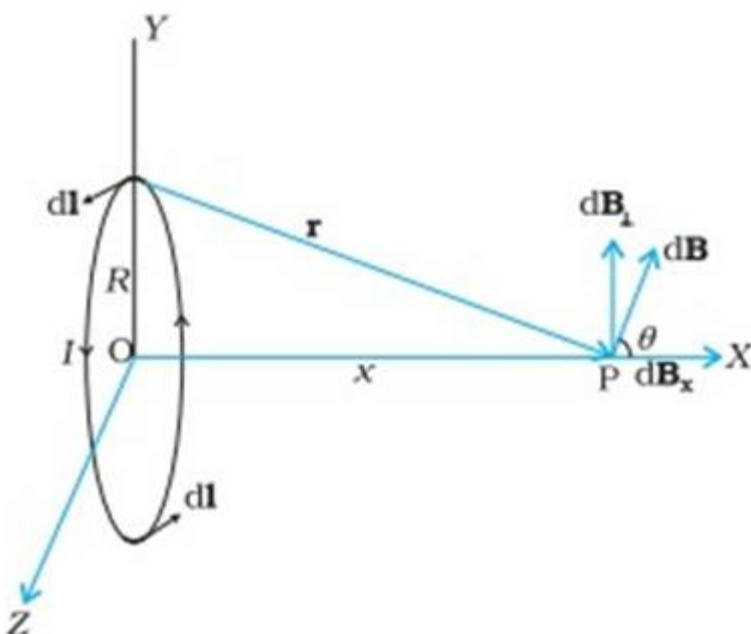
Question 29:

Write the formula of Biot-Sevart law in vector form. Obtain an expression of magnetic field on the axis of a current carrying circular loop. Draw necessary diagram.

Sol:

Biot-Sevart law can be written as,

$$d\vec{B} = \frac{\mu_0 i (d\vec{l} \times \vec{r})}{4\pi r^3}$$



Let us consider we have circular loop carrying a steady current I , placed in the y - z plane with its center at the origin O .

Let x be the distance of P from the center O . of the loop.

Now,

$$\begin{aligned} d\vec{B} &= \frac{\mu_0 i (d\vec{l} \times \vec{r})}{4\pi r^3} \\ &= \frac{\mu_0 I dl}{4\pi (x^2 + R^2)} \quad [\because r^2 = x^2 + R^2] \end{aligned}$$

Now,

$$\begin{aligned} \cos \theta &= \frac{R}{r} \\ \Rightarrow \cos \theta &= \frac{R}{\sqrt{x^2 + R^2}} \end{aligned}$$

Therefore, the magnetic field on the X -axis is:

$$dB_x = \frac{\mu_0 Idl}{4\pi(x^2 + R^2)} \times \frac{R}{\sqrt{x^2 + R^2}}$$

$$= \frac{\mu_0 Idl}{4\pi(x^2 + R^2)^{\frac{3}{2}}}$$

$$= \frac{\mu_0 IR^2}{2(x^2 + R^2)}$$

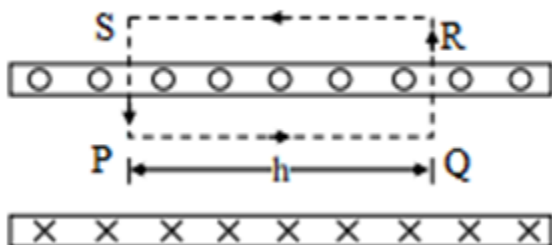
OR

Write Ampere's circuital law in mathematical form. Derive an expression of magnetic field at the axis of a current carrying long solenoid. Draw necessary diagram.

Sol:

The line integral $\oint \vec{B} \cdot d\vec{l}$ is μ_0 . (Permeability constant) times the net current I through the area bounded by the curve.

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I$$



Now, Apply Ampere's circuital law, for that let us choose a rectangular path PQRS

$$\oint \vec{B} \cdot d\vec{l} = \int_P^Q \vec{B} \cdot d\vec{l} + \int_Q^R \vec{B} \cdot d\vec{l} + \int_R^S \vec{B} \cdot d\vec{l} + \int_S^P \vec{B} \cdot d\vec{l}$$

$$\Rightarrow \mu_0 I = \int_P^Q B dl \cos \theta + \int_Q^R B dl \cos \theta + \int_R^S B dl \cos \theta + \int_S^P B dl \cos \theta$$

$$\Rightarrow \mu_0 I = \int_P^Q Bh \cos 0^\circ + \int_Q^R B dl \cos 90^\circ + \int_R^S (0) dl \cos 0^\circ + \int_S^P B dl \cos 90^\circ$$

$$\Rightarrow \mu_0 I = Bh$$

So, the net current I_0 which passes through the area bounded by the path of integration is not the same as the current I in the solenoid.

$$bh = \mu_0 Inh$$

$$\Rightarrow B = \mu_0 In$$

Question 30:

What is Polaroid? Write any two uses of it. How will you identify the polarised light, the partially polarized light and the unpolarised light?

Sol:

Large sized manufactured polarizing films capable of producing plane polarised beams of large cross-section is known as Polaroid.

Its uses are:

- (i) In sunglasses, windowpanes, etc.
- (ii) In photographic cameras and 3D movie cameras.

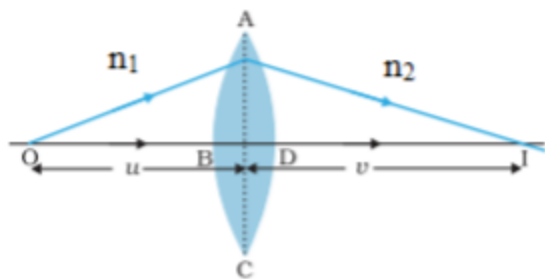
Polarized can be defined as the intensity of light after passing through a polarizer becomes two times maximum and two times minimum. While unpolarized light can be defined as the intensity of light after passing through a polarizer does not alter or changes. And the intensity of light after passing through a polarizer changes slightly, then it is a partially polarized light.

OR

Derive Lens maker formula $\frac{1}{f} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$. The lower half of the concave mirror's reflecting surface is covered with an opaque material. What will be the effect on the image formed by the mirror?

Sol:

Let us consider the thin lens shown in the image above with 2 refracting surfaces having the radii of curvatures R_1 and R_2 respectively.



Now, applying the formula of refraction at ABC and ADC we get,

$$\begin{aligned} \frac{n_2}{v_1} - \frac{n_1}{u} \\ = \frac{n_2 - n_1}{R_1} \dots\dots\dots(1) \end{aligned}$$

$$\begin{aligned} \frac{n_1}{v} - \frac{n_2}{v_1} \\ = \frac{n_1 - n_2}{R_2} \dots\dots(2) \end{aligned}$$

On adding eq(1) and eq (2) we get,

$$\begin{aligned} \frac{n_2}{v_1} - \frac{n_1}{u} + \frac{n_1}{v} - \frac{n_2}{v_1} &= (n_2 - n_1) \left[\frac{1}{R_2} - \frac{1}{R_1} \right] \\ \Rightarrow n_1 \left(\frac{1}{v} - \frac{1}{u} \right) &= (n_2 - n_1) \left[\frac{1}{R_2} - \frac{1}{R_1} \right] \\ \Rightarrow \left(\frac{1}{v} - \frac{1}{u} \right) &= \left(\frac{n_2 - n_1}{n_1} \right) \left[\frac{1}{R_2} - \frac{1}{R_1} \right] \\ \Rightarrow \frac{1}{f} &= (n - 1) \left[\frac{1}{R_2} - \frac{1}{R_1} \right] \quad \left\{ \because \frac{1}{f} = \frac{1}{v} - \frac{1}{u} \text{ and } n = \frac{n_2}{n_1} \right\} \end{aligned}$$